

Day 14 of training

React js

History

React was created by **Jordan walke** a software engg at facebook

It was first deployed for facebook feed around 2011

In May **2013**, **React** was Open sourced at js conference

what is React

React is a JavaScript library created for Fast and interactive user interface for web and mobile applications

React is a front-end JavaScript library in the field of web development

Popular websites Like Netflix,Airbnb,Yahoo!mail and many more use react to built there UI

OR

React is a tool used to build websites.

It helps you create parts of a website (like buttons, menus, forms) in a smart, reusable way using JavaScript.

Why React ?

Easy creation of dynamic application

.Reusable components

small learning cureve

unidirectional data flow

It can used for both mobile and web

dedicating tools for easy debugging

feature of javascript

JSX javascript syntax extension -JSX is a syntax extension to javascript .

it is used with react to describe what the user interface should look like

JSX lets you write HTML-like code in JavaScript — so you can design your UI and write logic in the same place.

Example:-

```
const element = <h1>Hello, Opinder!</h1>;
```

Virtual DOM – Easy Definition:

Virtual DOM is a lightweight copy of the real DOM used by React to make updates faster and smarter. Performance means how fast and smooth your website or app works — like how quickly it loads, updates, or responds to clicks

Debugging –

Debugging means finding and fixing errors (bugs) in your code so it works correctly.

Example:

If your button doesn't work, debugging is like:

Checking the code

Finding the typo or logic mistake

Fixing it

Testing again

Node.js – Easy Definition:

Node.js is a JavaScript runtime that lets you run JavaScript outside the browser — like on your computer or server.

Simple Words:

JavaScript usually runs in browsers (like Chrome).

Node.js lets JavaScript run on your machine like Python or C programs.

What is package.json?

It's a special file in every Node.js/React project that:

Stores project info (name, version, description).

Lists dependencies (what packages your project needs to work).

Defines scripts (shortcuts to run commands like npm start or npm run build).

Feature	`package.json`	`package-lock.json`
Purpose	Tells what packages your project needs	Locks exact versions of those packages
Who creates it?	You or tools like `npm init` by `npm install`	Automatically created
Version control	Uses version ranges (`^`, `~`, etc.) versions (`1.2.3`)	Uses exact
Editable?	Yes, you edit this file directly it	No, you usually don't touch
Level	Top-level dependencies only dependencies too	Includes nested (child)
Use case	For humans to understand and manage project dependencies	For npm to reproduce the same environment every time